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Prediction of user engagement with News using machine learning algorithms

BAKALÁŘSKÁ PRÁCE

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Autor: Albina Akhunova

Vedoucí bakalářské práce: doc. Ing. Tomáš Kliegr Ph.D.

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Abstrakt

V digitálním prostředí, kde je každý den publikováno velké množství zpravodajských článků, je porozumění faktorům ovlivňujícím uživatelskou interakci klíčové pro vydavatele, digitální platformy i inzerenty. Úroveň zapojení, vyjádřená například počtem komentářů, sdílení či kliknutí, odráží nejen atraktivitu obsahu, ale také jeho emocionální dopad a relevanci. Predikce tohoto zapojení na základě textových prvků umožňuje odhadnout potenciální úspěšnost článku ještě před jeho publikací. Současně napomáhá vydavatelům nalézt rovnováhu mezi tvorbou poutavého obsahu a zachováním důvěryhodnosti a odpovědnosti vůči uživatelům. Tato bakalářská práce se zaměřuje na predikci uživatelského zapojení s využitím metod strojového učení a zpracování přirozeného jazyka (NLP). Cílem je vytvořit model, který na základě textových prvků, zejména titulků a perexů, dokáže předpovědět míru očekávané interakce s článkem. Analýza vychází z článků publikovaných na portálu iDNES.cz. Práce kombinuje teoretický přehled současného výzkumu s praktickou implementací modelu strojového učení. Kromě návrhu a trénování modelu se práce věnuje také evaluaci výsledků a navrhuje možnosti dalšího rozvoje v oblasti predikce uživatelského zapojení.

Klíčová slova

Uživatelská Interakce, Zprávy, iDNES.cz, Strojové Učení, NLP

Abstract

In the digital environment, where a large number of news articles are published every day, understanding the factors influencing user engagement is crucial for publishers, digital platforms, and advertisers. The level of engagement, expressed for example through the number of comments, shares, or clicks, reflects not only the attractiveness of the content but also its emotional impact and relevance. Predicting this engagement based on textual features allows for estimating the potential success of an article even before its publication. At the same time, it helps publishers find a balance between creating compelling content and maintaining credibility and responsibility towards users. This bachelor's thesis focuses on predicting user engagement with news using machine learning methods and natural language processing (NLP) techniques. The goal is to develop a model that can predict the expected level of interaction with an article based on textual elements, particularly headlines and teasers. The analysis is based on articles published on the iDNES.cz portal. The thesis combines a theoretical overview of current research with the practical implementation of a machine learning model. In addition to the model's design and training, the work also evaluates its results and suggests possibilities for further development in the field of user engagement prediction.

Keywords

User Engagement, News, iDNES.cz, Machine Learning, NLP

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List of abbreviations

BERT	Bidirectional Encoder Representations from Transformers
LSTM	Long Short-Term Memory
NLP	Natural Language Processing
TF-IDF	Term Frequency-Inverse Document Frequency
BoW	Bag-of-Words
EDA	Exploratory Data Analysis
MAE	Mean Absolute Error
MSE	Mean Squared Error
RMSE	Root Mean Square Error
TP	True Positive
FP	False Positive
TN	True Negative
FN	False Negative

Introduction

In today's digital landscape, where thousands of news articles are published every day, understanding what drives users to engage with content is increasingly important for publishers, digital platforms, and advertisers. Engagement, often measured through metrics like comments, shares, and clicks, serves as a useful indicator of the relevance, interest, and emotional impact of an article. For media organizations, creating content that grabs attention is key to driving traffic, boosting revenue, and keeping the audience engaged over the long term.

However, while engagement plays a significant role in the financial success of a news portal, it is equally important for publishers to maintain trustworthiness and credibility with their users. Sensational or controversial content may generate more comments and shares, but it could also jeopardize the integrity of the news portal and lead to a decline in reader trust.

Predicting engagement is therefore crucial, as it can help publishers foresee and manage the balance between creating content that attracts attention and ensuring that content remains responsible and trustworthy.

The main goal of this bachelor thesis is to create a model that predicts user engagement with news articles, focusing on key textual features such as headlines and teasers. By applying machine learning techniques, the model will analyze how textual elements influence user interaction and classify articles based on their anticipated level of engagement. The experiment will be conducted on news articles from the IDNES.cz portal.

The motivation behind this bachelor thesis comes from a personal interest in machine learning and natural language processing. This project brings together both areas and offers an opportunity to explore how textual content can influence reader behavior, while also providing practical experience with the application of machine learning techniques.

Thesis structure

This bachelor thesis is divided into five chapters. The first chapter presents a literature review, discussing the concept of engagement, a framework for categorizing engagement metrics, and existing studies that utilize machine learning approaches for engagement prediction. The second chapter introduces machine learning, focusing on the various algorithms used in this field and the evaluation metrics employed to assess their performance. The third chapter is dedicated to Natural Language Processing, describing the techniques and methods used to process and represent textual data. The fourth chapter presents the experiment, detailing the methodologies and steps followed in the study. Finally, the fifth chapter focuses on the evaluation and comparison of the results. The thesis concludes with a summary of the findings and suggestions for future work.

1 Literature review

This chapter defines user engagement, introduces a framework for categorizing engagement metrics, and reviews studies that apply machine learning to predict and analyze user engagement with news articles.

1.1 User engagement and engagement framework

User engagement, as defined by Broersma (2019), *encompasses the cognitive, emotional, or affective experiences that users have with online content or brands* (s.1).

User engagement in the digital world is assessed through various metrics. These engagement metrics vary significantly in their importance, as they each capture distinct aspects of how users interact with digital content (Noguti, 2016). To better understand and categorize these metrics, Aldous et al. (2019) proposed an engagement framework. As shown in Table 1, the user engagement framework categorizes distinct levels of user interactions based on their public expressiveness (Aldous et al., 2019).

This framework builds electronic word-of-mouth and network effect theory, recognizing that the more publicly users engage with content, greater its potential to spread and influence others (Aldous et al., 2019).

Engagement level	Definition	Metric example
Level 1	Engaging with content in a private, non-expressive manner, meaning user's interaction is not visibly shared with others	Number of views, time on page, duration session, pages per session
Level 2	Users show more public interaction with online content by liking or favoriting.	Likes
Level 3	At this stage, users participate in the conversation, showing their opinions, emotions, or perspectives. Unlike simple reactions, these indicators	Comments, internal shares

	involve more effort and personal expression.	
Level 4	This is the most expressive and influential level of engagement, where users actively spread content beyond its original platform. These actions significantly enhance content reach.	External postings

Table 1 - User engagement framework (Source: Aldous et al., 2019)

1.2 Machine Learning in the field of engagement prediction

In this section, several use cases of machine learning applications in engagement prediction will be explored.

Balali et al. (2017) proposed a machine learning approach to predict the volume of comments on news articles from Iranian news portals, incorporating both textual and non-textual features. They focused on two classification problems: binary and multiclass. The binary model predicted whether an article would receive comments or not, dividing articles into two classes: With comments and Without comments. The multiclass model classified articles into three categories: No comment, Moderate comment, and High comment. To train these models, the authors extracted various features, including textual features such as unigram and bigram models to capture word occurrences, as well as temporal and global-textual features that considered the timing of the article's publication and overall text patterns. In addition, they incorporated various engagement metrics as non-textual features, such as the number of page visits, likes, and dislikes. The authors chose Random Forest classifier for both binary and multiclass problems. Their results showed high precision, recall, and F1-score for the binary classification task, achieving a weighted average of F1-score of 90.9%. For the multiclass task, the model demonstrated varying performance across the three categories, with the "High comment" category being predicted poorly. The weighted F1-score for the multiclass classification was 74.6%.

Cai et al. (2022) proposed a news popularity prediction model using a GRU-based deep neural network to predict the number of pageviews an article would receive after publication. The authors performed feature engineering, applying sentiment analysis and named entity extraction to the news headlines to create text features, while metadata features included news headline lengths, publishing time conversions, and author/category scores. The GRU model outperformed traditional methods like LSTM, linear regression, and random forest, achieving an MSE of 0.0152, MAE of 0.0865, and RMSE of 0.1098.

Hardt et al. (2017) explored predicting user engagement (i.e., article views) based on the text of online news articles. They investigated how much each text element-headline, teaser, and body contribute to predicting article popularity. The study utilized three classifiers: Support Vector Machines, Logistic Regression, and Random Forests, with Logistic Regression consistently outperforming the others in terms of accuracy and F1-score results. For representing textual data, authors employed methods like Bag-of-Words and word vectors. Their findings revealed that more text (combining headline, teaser, and body) improved predictive power of models. They also observed that text length and word vectors had a minimal impact on performance, while Bag-of-Words performing the best.

Lopez et al. (2022) conducted a study on predicting press articles virality by analyzing clicks as a key engagement metric. They aimed to determine which headlines would attract more user interactions using machine learning models. Authors extracted sentiments, verbs, adjectives, and named entities to enhance the model's understanding of linguistic patterns. They explored two main approaches: regression and classification. The regression models, including an LSTM-based architecture and a BERT-based ranking model, attempted to estimate the relative difference in clicks between two headlines, but failed to produce reliable results. The classification models, however, performed better, particularly the BERT-based classifier, achieved an accuracy of 64% in predicting which of two headlines would receive more clicks.

The study conducted by Tervonen et al. (2021) aimed to define and predict the effectiveness of online news headlines using machine learning algorithms. Headline effectiveness was measured by click-through rate and time spent on page. Based on these metrics, headlines were categorized into four classes: non-effective, appealing, engaging, and effective. To model headline effectiveness, the authors developed a deep learning classifier using BERT-based embeddings, comparing the performance of the multilingual BERT (multiBERT) with a Finnish-specific model (FinBERT). In addition to the contextual embeddings, they included manually extracted textual features (e.g., headline length, named entities), metadata, and article access policy (free vs. subscription) for feature engineering process. The results demonstrated that FinBERT significantly outperformed multiBERT in predicting headline effectiveness, emphasizing the advantage of language-specific models in domain-specific NLP applications. Manual features had minimal impact on performance, meaning that BERT embeddings effectively captured most of the relevant linguistic patterns.

2 Machine Learning

This chapter introduces Machine Learning. It explains various machine learning algorithms, describing how they work and where they are applied. The chapter concludes with a description of the evaluation metrics used to assess model performance.

2.1 Machine Learning and its tasks

Machine Learning can be defined as a part of Artificial Intelligence that allows computers to learn from data, identify patterns and relationships, and improve their performance on tasks without explicit programming (Samuel, 1959).

2.1.1 Supervised Machine Learning

The objective of supervised machine learning is to train an algorithm on a dataset where each input is matched with the corresponding to it labels. Once the training of a model is complete, it applies what it has learned to make predictions on new, unseen data.

Supervised learning focuses on two main problems: regression and classification. Classification tasks can be further divided into binary and multi-class problems. An example of a binary classification problem is predicting whether a person has diabetes, where the model distinguishes between two classes: "has diabetes" or "does not have diabetes". A multi-class classification problem, on the other hand, might involve tasks like image recognition, where the model categorizes pictures of animals into different classes.

The goal of regression is to predict continuous numerical values. For example, this could involve predicting a person's annual income based on features such as age, education level, occupation, marital status (Müller & Guido, 2016).

2.1.2 Unsupervised Machine Learning

In unsupervised learning, the model is given an unlabeled dataset and must identify patterns and connections within the data itself (Müller & Guido, 2016). Unsupervised learning solves two main types of problems: clustering and association rule mining.

Clustering is the process of partitioning data points into distinct groups based on their similarities (Müller & Guido, 2016). The algorithm analyzes the data and forms clusters by grouping similar points together without any predefined labels. A well-known example of this is how photo management software, like iPhoto, detects faces in pictures and automatically groups them into albums based on facial features, without prior identification (Apple Machine Learning Research, 2021).

Association rule mining focuses on discovering patterns in data by analyzing how different items or events are connected. A well-known example is market basket analysis, where businesses analyze customer purchase behavior to uncover patterns, such as the tendency for customers who buy bread to also purchase butter.

2.2 Machine Learning approaches

In this chapter, commonly used supervised machine learning algorithms are introduced, all of which were applied in this study for prediction purposes. Each subsection briefly outlines the core principles of the selected models, including Decision Trees, Random Forest, Support Vector Machine, Gradient Boosting, and Logistic Regression.

2.2.1 Decision Trees

Decision Trees is a supervised machine learning algorithm, where the main idea is making a decision based on a sequence of if/else questions (Müller & Guido, 2016).

A decision tree grows from a root node, which is the starting point of the algorithm. It performs the initial split based on one of the dataset's attributes. The internal nodes represent questions or conditions that further split the data. Each internal node divides the dataset into subsets, according to the values of the selected by the algorithm attribute (Rokach & Maimon, 2005). Edges are the connections between the nodes, representing the paths that data follows based on the decisions made at each internal node. The tree ends with the leaf nodes, which contain the final answer (Rokach & Maimon, 2005).

ID3 is a decision tree algorithm that builds the tree based on Information Gain. Information Gain is derived from a metric called entropy, which measures the uncertainty in the data (Quinlan, 1986). ID3 constructs the tree by selecting the attribute that results in the greatest reduction in entropy, or the highest Information Gain, to make the decision at each node (Quinlan, 1986). This process is applied recursively to the resulting data subsets, gradually expanding the tree structure in a top-down manner (Quinlan, 1986).

2.2.2 Random Forests

A frequent problem with individual decision trees is overfitting, which causes the model to struggle in generalizing to unseen data (Breiman, 2001). Random Forests algorithm overcomes this by constructing a collection of independent decision trees. To build each tree, a bootstrap sample of the data is taken, meaning data points are randomly selected from the original dataset with replacement (Breiman, 2001). To ensure variability across the trees, a random subset of features is considered at each split. Although not every tree needs to be highly accurate, the strength of the model lies in the collective decision-making of a large number of trees. For final prediction, each decision tree casts a vote, and the Random Forests selects the option that receives the majority of votes (Breiman, 2001).

2.2.3 Gradient Boosting

Gradient Boosting is a machine learning algorithm that combines simple models, typically shallow decision trees, and makes predictions by aggregating them into a powerful final model (Friedman, 2001). Each new simple model is trained to predict the pseudo-residuals, which are the differences between the actual target values and the predictions made by the current model. The core idea behind this algorithm is to minimize these residuals, progressively improving the model's performance (Friedman, 2001).

Gradient Boosting starts with an initial, often simple, model denoted as $F_0(x)$, which is usually the mean of the target values (Friedman, 2001). At each iteration m , the model is updated by training a new weak learner $h_m(x)$ to approximate the pseudo-residuals $r_i^{(m)}$, which represent the negative gradient of the loss function $L(y_i, F(x_i))$ with respect to the current model's predictions (Friedman, 2001). The model from the previous iteration, $F_{m-1}(x)$ is then updated by adding the new weak learner scaled by a learning rate v resulting in a new model $F_m(x)$ (Friedman, 2001).

2.2.4 Support Vector Machines

Support Vector Machines aim to classify data points by finding the optimal decision boundary that most effectively separates classes within a feature space (Vapnik & Cortes, 1995). This boundary is selected to maximize the distance between it and the closest data points from each class (Vapnik & Cortes, 1995).

There are two main variants of Support Vector Machines: the hard margin SVM and the soft margin SVM. The hard margin approach assumes that the data is smoothly separable by a straight line, meaning that the data points from two distinct classes do not overlap with each other (Vapnik & Cortes, 1995). It seeks the maximum-margin decision boundary under the strict condition that all data points are correctly classified and lie outside the margin. In contrast, the soft margin SVM allows a certain degree of misclassification to effectively deal with non-linearly separable data (Vapnik & Cortes, 1995). This flexibility is controlled through a regularization parameter, which balances the trade-off between maximizing the margin and minimizing classification error (Vapnik & Cortes, 1995).

2.2.5 Logistic regression

Logistic regression is a supervised learning algorithm applied both to binary and multi-class classification tasks. It derives its name from the fact that it employs the logistic (sigmoid) function to predict the probability that a given data point should be classified under a particular class (Bisong, 2019). Logistic regression works by computing a linear combination of the input features and their corresponding weights, which defines a linear hyperplane in the feature space (Bisong, 2019). This result is then passed through the sigmoid function, which transforms the linear output into a probability indicating the likelihood that the instance belongs to a specific class (Bisong, 2019).

In the logistic regression model, the probability that the target variable Y belongs to class 1 given predictors X is defined as:

$$P(Y = 1 | X) = \frac{1}{1 + e^{-\beta_0 - \beta_1 X_1 - \beta_2 X_2 - \dots - \beta_p X_p}}$$

where:

- β_0 is the intercept,
- β_i (for $i = 1, \dots, p$) are the coefficients associated with each feature X_i ,
- X_i are the predictor variables.

2.3 Evaluation metrics

The following sections introduce the primary evaluation metrics relevant to various machine learning scenarios.

2.3.1 Confusion Matrix

Confusion matrix displays the outcomes of model predictions by showing how many instances were correctly or incorrectly classified. The matrix is composed of four categories: True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) (Varoquaux & Colliot, 2023).

The figure shown below represents a 2x2 confusion matrix, which is typically used for a binary classification type of problem. The TP in the top-left cell represents correctly classified positive class instances. The FP, found in the top-right cell, indicates cases where the model incorrectly predicted actual negative class instances as positive ones. The FN in the bottom-left cell reflects instances where the model mistook actual positive class instances as negative ones. The TN in the bottom-right cell represents instances where the model correctly predicted the negative class.

		True Label	
		Positive	Negative
Predicted Label	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

Figure 1 - Confusion matrix (Source: author)

2.3.2 Accuracy

Accuracy is one of the most fundamental evaluation metrics for classification models, representing the proportion of correctly predicted observations out of all observations (Grandini et al, 2020). However, it is important to note that accuracy tends to be biased towards the majority class. In scenarios where the dataset is imbalanced, a model can achieve a high accuracy score by predominantly predicting the majority class, even if it fails to properly classify the minority class (Grandini et al, 2020). As a result, accuracy is not a reliable metric when working with imbalanced datasets, and alternative evaluation metric should be considered.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

2.3.3 Precision

Precision is the proportion of correctly classified positive observations to all positively classified observations (Vakili et al., 2020). This metric is especially important in contexts where false positives could lead to significant consequences, for example, in disease prediction, where incorrect positive results can lead to unnecessary treatment or interventions.

$$\text{Precision} = \frac{TP}{TP + FP}$$

2.3.4 Recall

Recall is a metric that measures the proportion of true positive instances correctly identified by the model out of all the actual positive cases (Vakili et al., 2020). It evaluates how well the model captures all the positive instances, specifically focusing on minimizing false negatives, instances that the model wrongly classifies as negative when they are actually positive (Grandini et al, 2020). This is particularly important in tasks where missing positive observations can be critical, such as in fraud detection.

$$\text{Recall} = \frac{TP}{TP + FN}$$

2.3.5 F1-Score

The F1-score is a harmonic mean of precision and recall; it offers a single metric that reflects both false positives and false negatives (Grandini et al, 2020). Additionally, the F1-score is an effective metric when dealing with imbalanced class distributions, as it gives a more comprehensive view of model performance than accuracy alone.

$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

3 Natural Language Processing

Since the core of this work involves textual data, it is important to provide an overview of the Natural Language Processing techniques relevant to the experiment ahead. This chapter will highlight selected methods for processing and representing text in ways that make it suitable for modeling and prediction.

3.1 Text Preprocessing

This subchapter introduces the basic techniques of text preprocessing that are critical for preparing text data for analysis and modeling.

3.1.1 Text cleaning

Text preprocessing usually starts with lowercasing (Egger & Gokce, 2022). It ensures that words are not treated as distinct due to differences in capitalization, for example, "Zpravodaj" and "zpravodaj" would otherwise be counted separately (Egger & Gokce, 2022). Text cleaning also involves the removal of punctuation, numbers, emojis, and URLs. This process eliminates irrelevant characters and unnecessary noise, so that the meaning behind the text becomes clearer and the data is better prepared for further analysis (Egger & Gokce, 2022).

3.1.2 Tokenization

Tokenization is the process of splitting a text document into individual components, known as tokens (Egger & Gokce, 2022). These components may include words, sentences, or other meaningful units. Tokenization in English is relatively straightforward, as words are generally separated by spaces and punctuation. However, tokenization in Czech is more complex due to the language's rich morphology, which results in numerous word forms and compound words (Zandl, 2024). Additionally, Czech includes diacritics that can change word meanings, adding another layer of complexity to the tokenization process (Zandl, 2024). Therefore, tokenization for Czech requires more sophisticated methods to ensure accurate word segmentation and handling of unique language features.

3.1.3 Stop words removal

Stop words typically serve grammatical purposes rather than adding substantial content to a sentence. Examples include conjunctions, prepositions, and other short functional words (Egger & Gokce, 2022). Because of their frequent occurrence, stop words can be safely removed without losing key information, allowing to focus on the more significant words that carry the core meaning of the text (Egger & Gokce, 2022).

3.1.4 Lemmatization and stemming

Lemmatization and stemming are two common text preprocessing techniques used to reduce word forms to a shared base, thereby minimizing redundancy in the vocabulary (Egger & Gokce, 2022). Stemming operates by cutting off endings, prefixes, suffixes, to reach a root form, which is often not a valid word (e.g., *dělali* → *děl*), whereas lemmatization relies on linguistic rules and considers the part of speech to produce dictionary-based lemmas (e.g., *dělali* → *dělat*) (Egger & Gokce, 2022). In languages with rich inflection, like Czech, lemmatization is generally preferred over stemming (Seznam.cz, 2011).

3.2 Text Representation

Since machine learning models require numerical inputs, textual data must be carefully preprocessed and transformed into a structured numerical format. Text representation techniques like Bag-of-Words and Term Frequency-Inverse Document Frequency serve this purpose by converting text into vectors based on word frequency.

3.2.1 TF-IDF

TF-IDF is a text representation method that measures the significance of a word in a particular document relative to the corpus (Egger, 2022). Term Frequency quantifies how frequently a word appears in a document relative to the total number of words in that document (Egger, 2022). On the other hand, Inverse Document Frequency assesses the rarity of a word across the entire corpus (Egger, 2022). IDF helps reduce the influence of commonly used words such as conjunctions or prepositions that appear frequently but carry little meaningful information. This way, words that occur in many documents receive lower scores, while those that are more specific to fewer documents are given greater importance.

$$\text{TF}(t, d) = \frac{f_{t,d}}{\sum_k f_{k,d}}$$

$$\text{IDF}(t) = \log\left(\frac{N}{n_t + 1}\right)$$

$$\text{TF-IDF}(t, d) = \text{TF}(t, d) \cdot \text{IDF}(t)$$

where:

- t represents a term (word) in the corpus,
- d represents a document in the corpus,
- $f_{t,d}$ represents how many times the term t appears in document d ,
- $f_{k,d}$ refers to the frequency of any term k in document d , where k could be any word in the document,
- N represents the total number of documents,
- n_t refers to the number of documents in the corpus that contain a term t .

3.2.2 Bag-of-Words

The Bag-of-Words (BoW) text representation technique transforms a document into a vector of fixed length, where each element indicates the frequency of occurrence of a specific word from the corpus vocabulary in the document. (Egger, 2022). BoW treats text as an unordered collection of words, meaning that it does not account for syntax and grammar. As a result, the context in which words appear is ignored, and the model only captures the frequency of individual words (Egger, 2022).

4 Experiment Flow

This chapter outlines the experiment flow. It describes the data collection process, dataset characteristics, exploratory data analysis, tools and libraries utilized, and the overall prediction process, including preprocessing, feature engineering, data splitting, and the selection of machine learning models.

4.1 Data collection

The primary data source for this research is iDNES.cz (available at <https://www.idnes.cz/>), a prominent and widely read online news portal in the Czech Republic. As direct automated web scraping of the portal is not allowed due to its terms of service and technical limitations, the dataset was obtained with the assistance of the portal's IT administrator. The dataset encompasses articles published over a one-year period, from January 1, 2024, to December 31, 2024. It was provided in Comma-Separated Values (CSV) format, a widely accepted and efficient format for data analysis.

4.2 Dataset

The dataset contains 13,212 records and consists of five columns: article category, headline, teaser, publication date, and number of comments. Since the only engagement metrics available on the IDNES.cz portal for an article is either the number of shares or the number of comments, for this study, the number of comments was selected as the primary engagement metric. This metric serves as the basis for evaluating user interaction and will be used to derive the categorical engagement levels (specifically, Low and High) that the machine learning models will aim to predict. The choice to use comments is justified as they represent a more direct and active form of user interaction and discussion sparked by the article's content. The textual data (headline, teaser) is primarily in the Czech language.

Table 2 provides an overview of the dataset used in this study.

Column name	Data type	Description
article_category	text	The category or section of the article (zpravodaj)
headline	text	The name of the article
teaser	text	Preview of the article

publication_date	datetime	The date and time the article was published
comments_count	integer	The number of comments the article has received.

Table 2 - Dataset description (Source: author)

4.3 Problem Type: Classification vs. Regression

The original task of this thesis involved predicting the exact number of comments using regression models. However, the distribution of the target variable was highly skewed, with most articles receiving very few comments and a small number receiving a disproportionately high number of comments. This imbalance made it difficult for the models to learn accurate patterns. To address this, a log-transformation was applied to reduce skewness, but the models continued to perform poorly, showing high prediction errors (e.g., unsatisfactory MAE values). Another challenge was that the key textual features, such as headlines and teasers, were written in Czech, which complicated preprocessing and limited the performance of standard NLP techniques. Due to these issues, the problem was reformulated as a binary classification task, where articles were categorized into two engagement levels: Low and High, based on the number of comments. Using the same preprocessing and feature engineering pipeline, the classification models achieved significantly better performance than the regression ones, meaning that the available features are more effective for distinguishing engagement levels rather than predicting exact number of comments.

4.4 Exploratory Data Analysis

In this section, exploratory data analysis (EDA) is performed to better understand the dataset. Visualizations and summary statistics are used to explore the structure and distribution of the data, which can help in further feature engineering and model development.

4.4.1 Comments count

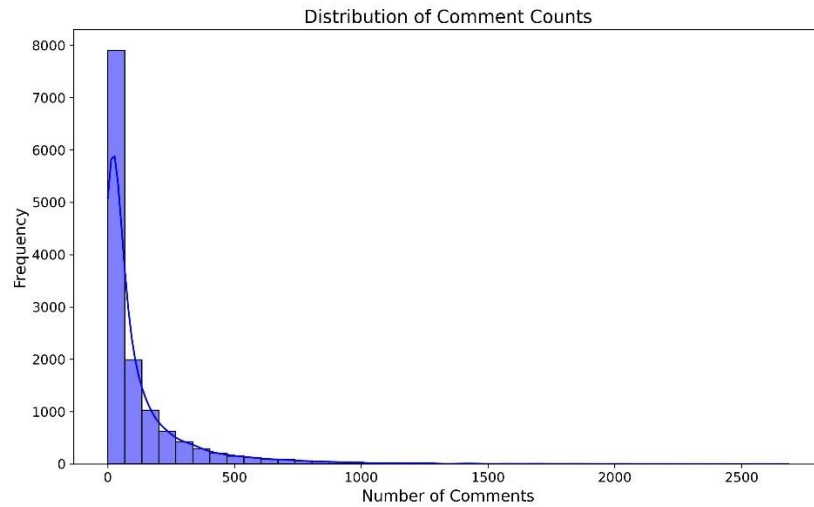


Figure 2 - Comments count histogram (Source: author)

Figure 2 shows a histogram of comment counts that reveals a right-skewed distribution, where most articles receive a low number of comments, while a smaller subset gets significantly higher engagement. To reduce the influence of these extreme values and better prepare the data for modeling, a logarithmic transformation was applied to the comment counts. This transformation helps to normalize the distribution and makes it more suitable for further analysis.

To form the classification target for engagement prediction, the threshold for the engagement level is calculated by determining the 75th percentile of the log-transformed comments count. This threshold is used to distinguish between high and low engagement levels. If a value in logarithmic transformed comment count is greater than or equal to the threshold, the engagement level is labeled as "High Engagement"; otherwise, it is labeled as "Low Engagement". This binary classification target is then used to predict whether an article will have high or low engagement based on the available features.

4.4.2 Engagement level (target)

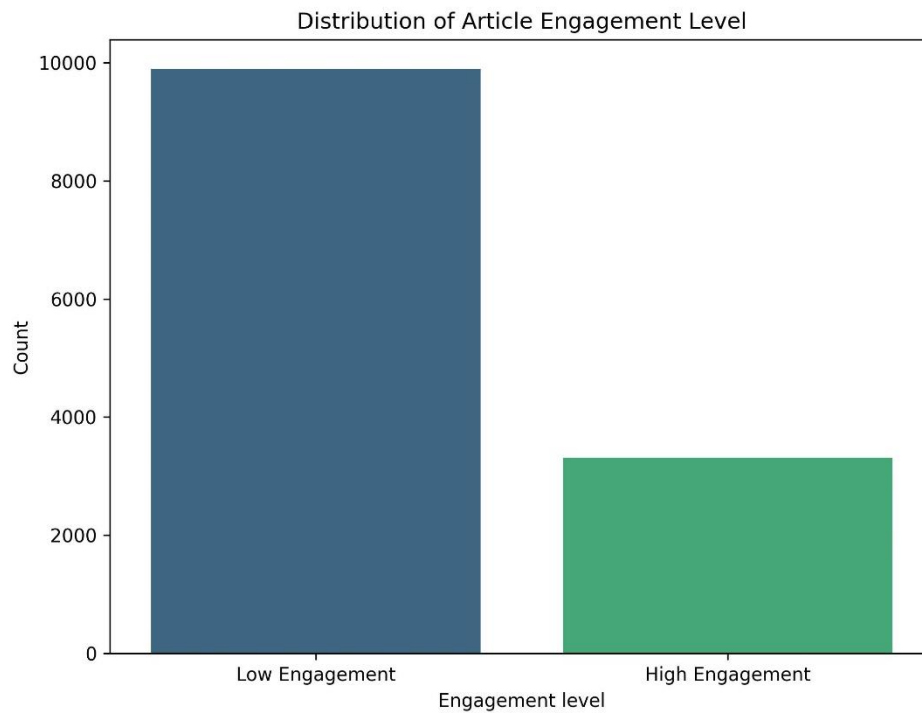


Figure 3 - Target distribution (Source: author)

Figure 3 presents the distribution of the derived binary target variable, 'Engagement Level'. A significant class imbalance is evident, with the 'Low Engagement' category containing most articles compared to the 'High Engagement' category. This imbalance requires using appropriate techniques during model training, as well as choosing evaluation metrics that provide reliable results for imbalanced data.

4.4.3 Box Plots of Comment Counts by Engagement Level

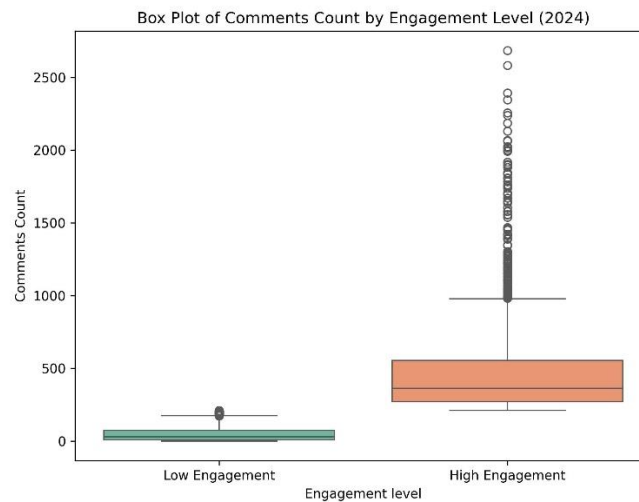


Figure 4 - Box Plots of Comment Counts by Engagement Level (Source: author)

The boxplots in Figure 4 show a clear difference in the distribution of comment counts between Low and High Engagement articles. Low engagement articles have few comments and limited variation, while high engagement articles show a much wider spread and many high-value outliers. These outliers likely reflect truly popular or controversial content, not errors. Removing them would hide important patterns in peak engagement, so they are kept in the dataset to capture the full range of user interaction.

4.4.4 Word Clouds for Headline and Teaser



Figure 5 - Word Cloud for headline (Source: author)

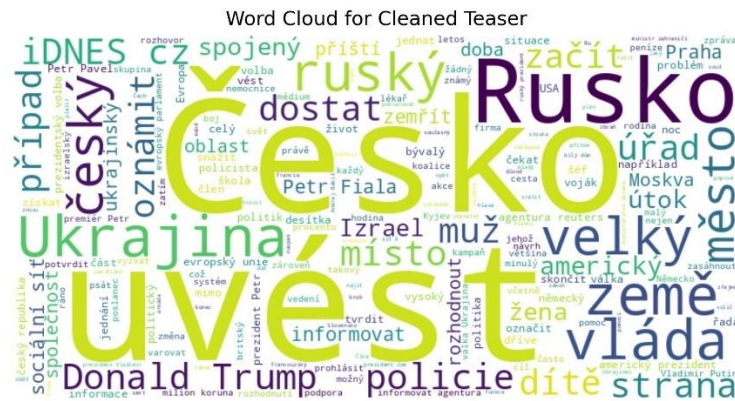


Figure 6 - Word Cloud for teaser (Source: author)

Figures 5 and 6 show word clouds generated from the processed headlines and teasers, providing a clear visualization of the most frequent terms in each text type. Both highlight a strong focus on themes such as geopolitics, national identity, and emotionally charged topics. Terms like "Rusko" (Russia), "Ukrajina" (Ukraine), "Izrael" (Israel), and "Trump" signal international conflicts, while words such as "vláda" (government) and "policie" (police) reflect national political issues. Additionally, emotionally impactful terms like "dítě" (child), "útok" (attack), and "válka" (war) are likely to provoke stronger emotional reactions.

4.5 Tools and libraries

For the execution of this experiment, a variety of Python packages were used to implement machine learning models, preprocess textual data, and perform visualizations. Below is a brief description of the packages and tools utilized in this experiment:

- **Python** is an interpreted, interactive, object-oriented programming language known for its clear syntax and versatility (Python, 2025).
- **Google Colab** is a cloud-based platform that provides an interactive Jupyter Notebook environment, allowing users to run Python code without any setup (Bisong, 2019).
- **Pandas** is a Python library designed for data manipulation, offering powerful tools to work with structured data in different formats (McKinney, 2010).
- **Numpy** is a Python library that provides efficient tools for working with arrays and performing numerical operations (Harris et al., 2020).
- **Matplotlib** is a Python library used for creating a wide range of visualizations for data analysis purposes (Hunter, 2007).

- **Transformers** is a library developed by Hugging Face that offers a wide range of pre-trained models for natural language processing tasks, including text classification, sentiment analysis, and more (Wolf et al., 2020).
- **Scikit-learn** is a Python library that offers simple and efficient tools for machine learning tasks such as classification, regression, clustering, and model selection (Pedregosa et al., 2011).
- **Stanza** is a natural language processing library developed by the Stanford NLP Group, offering tools such as tokenization, lemmatization, and dependency parsing, and was chosen for this project due to its support for the Czech language (Qi et al., 2020).

4.6 Prediction Flow

This subchapter describes overall prediction process, including preprocessing, feature engineering, data splitting, and the selection of machine learning models.

4.6.1 Preprocessing and Feature Engineering

Since the primary features used in engagement prediction are textual features, it is important to carefully preprocess them. Given that the dataset consists of Czech news, lemmatization was chosen instead of stemming, as Czech is a highly inflected language, and lemmatization helps normalize different word forms. To preprocess the headline and teaser, the **Stanza** library was used, as it effectively tokenizes and lemmatizes Czech text.

Because popular libraries such as **NLTK** do not support Czech stopwords, a predefined list from an external **repository**¹ named **stop-words** was used instead. This stopword list is easily accessible and can be conveniently installed through pip. However, the stopword list from the **stop-words** repository was incomplete and missed some common Czech words, including frequent prepositions and short function words like "ze", "s", "v", and "z". To identify additional missing words, word clouds were analyzed, which revealed high occurrences of such very short, common words. Furthermore, a simple regular expression function was used to extract all one- and two-letter words from the dataset, helping systematically find and review frequent short tokens that were not initially filtered out. The text was also cleaned from noise like punctuation and numbers. Speaking of lowercasing, Stanza automatically lowercases most words while preserving proper casing for named entities such as names, organizations, or locations.

¹ <https://github.com/Alir3z4/stop-words>

Tables 3 and 4 illustrate examples of raw and cleaned text for the headline and teaser, respectively. These examples demonstrate how the preprocessing steps simplify the text while retaining essential information.

headline	cleaned_headline
Dilema lesů: kůrovec, nebo sucho? Sázání nových stromů může být riziko	dilema les kůrovec sucho sázání strom riziko

Table 3 - Raw vs cleaned headline (Source: author)

teaser	cleaned_teaser
České lesy mění tvář. Do smrkových monokultur se vysazuje čím dál víc listnáčů, zejména buků. Vědci však přišli s analýzou, že buky vysušují půdu víc než smrky. Co s tím?	český les měnit tvář smrkový monokultura vysazovat listnáč zejména buk vědec analýza buk vysušovat půda smrka

Table 4 - Raw vs cleaned teaser (Source: author)

Then, the cleaned headline and teaser were combined into a single column. This combination was done to provide a more unified context, as both the headline and teaser together serve as a comprehensive summary or preview of the article. To transform the combined text into numeric form, TF-IDF text representation technique was applied, capturing the most important terms while accounting for their frequency and relevance across the entire dataset. To reduce dimensionality the number of terms was reduced to 5,000. The limit of 5,000 terms for the TF-IDF vectorization was chosen as a balance between dimensionality reduction and preserving enough information from the text. This choice balanced retaining important features while ensuring computational efficiency. By limiting the feature space, the model complexity was kept manageable, helping to avoid overfitting and reduce training time.

After preprocessing, feature engineering was performed to capture temporal patterns that could influence engagement. This included extracting the publication hour, day of the week, and month from the timestamp.

To complement the textual content with emotional context, sentiment analysis was applied to both the headline and teaser using a multilingual **RoBERTa-based model**². The model was trained on a large multilingual corpus, with approximately 100,000 tweets in Czech, placing it within the top 30 most frequent languages in the dataset (Barbieri et al., 2022). It classifies text into three sentiment categories: positive, neutral, and negative. These

² <https://huggingface.co/cardiffnlp/twitter-xlm-roberta-base-sentiment>

sentiment labels were subsequently one-hot encoded to be used as input features in downstream modeling.

4.6.2 Data Splitting

The data was split into training, validation, and test sets with a 60/20/20 split. This division was chosen because the models will be trained with different hyperparameters, and the validation set is necessary for hyperparameter tuning. **GridSearchCV**³ was applied to all models for hyperparameter tuning.

4.6.3 Machine Learning Models

In this subchapter, the machine learning algorithms selected for the task of engagement prediction are presented.

1. Logistic Regression

Logistic Regression was chosen for engagement prediction due to its simplicity, efficiency, and effectiveness in binary classification. Given the class imbalance in the dataset (with more low engagement articles than high engagement ones), Logistic Regression was further adjusted using the **class_weight='balanced'** parameter. This adjustment allows the model to penalize the majority class, helping it focus more on the minority class (high engagement articles). While various techniques, including random oversampling, were tested, it was observed that Logistic Regression with oversampling led to overfitting, with the model performing exceptionally well on the training set but poorly on the validation and test sets. In contrast, Logistic Regression with the **class_weight='balanced'** parameter showed more consistent results and did not suffer from overfitting. During internal experiments, two strategies were compared: **random oversampling** and **class_weight** balancing. When random oversampling was applied, the model achieved an extremely high training accuracy of 99.01%, but validation and test accuracies dropped significantly to 71.62% and 70.60%, respectively, which showed strong overfitting. Moreover, the F1-score for the minority class (high engagement) was only 0.42. In contrast, using the **class_weight='balanced'** parameter resulted in a lower but more realistic training accuracy of 83.55%, with validation and test accuracies of 73.67% and 72.00%. The F1-score for the minority class improved to 0.51.

Table 5 shows the hyperparameters tested for Logistic Regression.

³ https://scikit-learn.org/stable/modules/generated/sklearn.model_selection.GridSearchCV.html

Hyperparameter	Values Tested	Best Value
C	[0.01, 0.1, 1, 10, 100]	1
solver	['lbfgs', 'liblinear']	'liblinear'
penalty	['l2']	'l2'

Table 5 - Hyperparameters for Logistic Regression (Source: author)

2. Decision Trees

This model was selected because it is easy to understand and interpret. Although it doesn't offer the performance gains of more advanced ensemble methods like Gradient Boosting or XGBoost, it serves as a useful point of comparison. Like logistic regression, two approaches were tested to handle the class imbalance: **random oversampling** and adjusting the **class_weight** parameter. The oversampling approach led to overfitting, with the model achieving 100% accuracy on the training set but only 66.10% test accuracy and a F1-score of 0.39 for the minority class (high engagement). In contrast, using **class_weight='balanced'** resulted in a 76.43% training accuracy, 71.40% test accuracy, and an improved F1-score of 0.45 for the minority class. As a result, the class-weighted decision tree model was selected for final evaluation, as it better handled the class imbalance and generalization.

Table 6 shows the hyperparameters tested for Decision Tree Classifier.

Hyperparameter	Values Tested	Best Value
max_depth	[None, 10, 20]	10
min_samples_split	[2, 5, 10]	2
min_samples_leaf	[1, 5, 10]	10

Table 6 - Hyperparameters for Decision Tree Classifier (Source: author)

3. Random Forest

Random Forest was selected for engagement prediction due to its robustness and strong performance with high-dimensional datasets. Another advantage is its ability to provide feature importance scores, which tell what features most contribute to the model's predictions. To address the class imbalance, two internal experiments were conducted: random oversampling and adjusting the **class_weight='balanced'**

parameter. With **random oversampling**, the model achieved a minority class F1-score of 0.50 on the test set, which was the same as the **class_weight='balanced'** approach, which also resulted in an F1-score of 0.50 for the minority class. However, the **class_weight='balanced'** approach outperformed the oversampling approach in terms of recall (0.55 for **class_weight='balanced'** and 0.53 for oversampling), which is crucial for detecting the minority class. Therefore, the Random Forest model trained with **class_weight='balanced'** was selected for the final evaluation.

Table 7 shows the hyperparameters tested for Random Forest.

Hyperparameter	Values Tested	Best Value
n_estimators	[100, 150]	150
max_depth	[10, 20]	20
min_samples_split	[5, 10]	5
min_samples_leaf	[5, 10]	5
max_features	['sqrt']	'sqrt'

Table 7 - Hyperparameters for Random Forest Classifier (Source: author)

4. SVM

SVM was selected for engagement prediction due to its effectiveness in finding an optimal separating hyperplane, especially in high-dimensional spaces like the TF-IDF representation of textual features. SVM is well-suited for binary classification problems and tends to perform well when the data is not perfectly linearly separable by maximizing the margin between classes. As with the other models used, the SVM was configured with **class_weight='balanced'** to directly address the class imbalance during training. This approach, again, proved to be more effective than applying oversampling techniques. Internal experiments comparing **class_weight = 'balanced'** against random oversampling confirmed this decision: using **class_weight = 'balanced'** resulted in a minority class (high engagement) F1-score of 0.51, whereas random oversampling achieved only 0.41. Moreover, the oversampled model exhibited strong overfitting with a training accuracy of 97.79% but a lower test accuracy of 69.05%, while the **class_weight='balanced'** approach demonstrated a more stable performance with a test accuracy of 72.95% and a training accuracy of 77.86%, offering a better balance between performance on the training and test sets.

Table 8 shows the hyperparameters tested for SVM classifier.

Hyperparameter	Values Tested	Best Value
C	[0.1, 1, 10]	0.1
Kernel	['linear', 'rbf']	'linear'
Gamma	['scale', 'auto']	'scale'

Table 8 - Hyperparameters for SVM classifier (Source: author)

5. XGBoost

In this study, XGBoost (Extreme Gradient Boosting) was selected as one of the primary classification models due to its superior performance in structured data problems and its robustness in handling various data complexities. XGBoost is an optimized implementation of the gradient boosting framework that has consistently achieved state-of-the-art results in machine learning competitions and practical applications. To address the class imbalance, Random Oversampling was applied to the training set using the **RandomOverSampler**⁴ method from the imblearn library. Initially, the training set consisted of 5936 samples of the majority class (low engagement) and 1990 samples of the minority class (high engagement). After oversampling, the minority class was randomly duplicated to match the majority class, resulting in a balanced training set with 5936 samples in each class. This approach was chosen to prevent the model from being biased towards the majority class without introducing synthetic data points, and to improve the model's ability to correctly predict minority class instances.

Table 9 shows the hyperparameters tested for XGBoost.

Hyperparameter	Values Tested	Best Value
n_estimators	[50, 100]	100
max_depth	[3, 5]	5
colsample_bytree	[0.8, 1]	0.8
subsample	[0.8, 1]	1
learning_rate	[0.01, 0.1]	0.1

Table 9 - Hyperparameters for XGBoost (Source: author)

4.6.4 Model interpretation with SHAP plot

While feature importance metrics, such as those from Random Forest, indicate which features are influential overall, they do not specify how these features contribute to individual class predictions. **SHAP**⁵ (SHapley Additive exPlanations) changes this by showing how specific features affect the model's predictions for each class.

Figure 7, shown below, illustrates the application of SHAP values to an XGBoost model.

⁴ https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.RandomOverSampler.html

⁵ <https://shap.readthedocs.io/en/latest/>

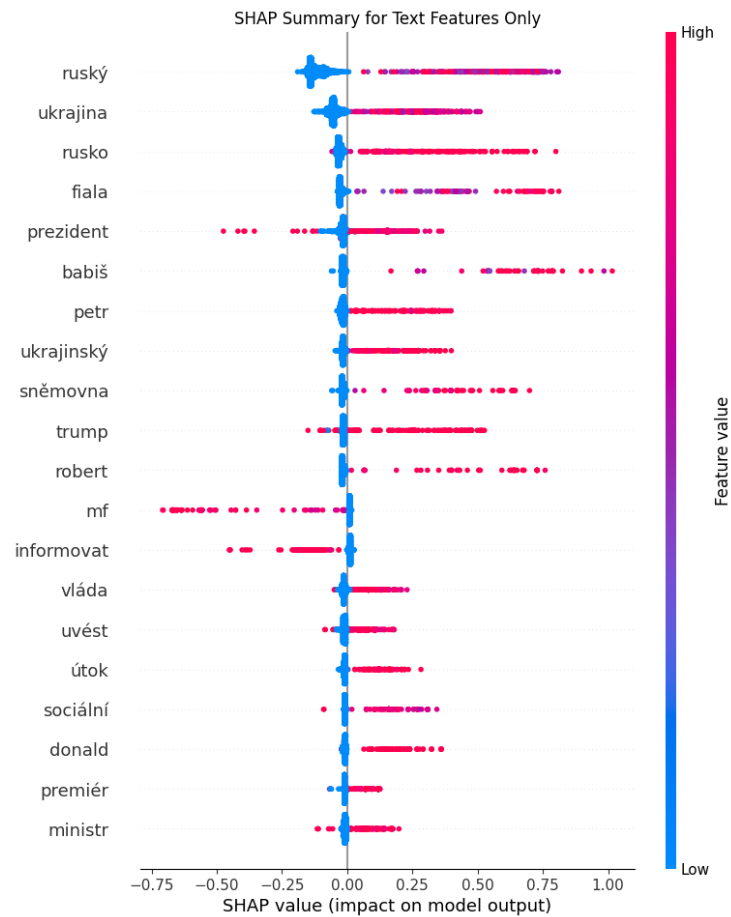


Figure 7 - SHAP plot (Source: author)

The results reveal that words associated with **geopolitical conflicts**, such as *ukrajina* (Ukraine), *rusko* (Russia) are strongly linked to higher engagement levels. Their presence in the articles significantly increases the likelihood of the model predicting high engagement. Similarly, the names of key **Czech political figures**, including *petr*, *fiala*, and *babiř*, along with references to political institutions such as *sněmovna* (the Chamber of Deputies), also contribute to high level engagement.

5 Evaluation and interpretation of the results

This chapter presents the evaluation of the classification models, performed on the test set, selected and implemented in this study. It also shows a comparison of the results between these classifiers. While other relevant evaluation metrics will be considered, given the imbalanced nature of the dataset, the F1-score will be a crucial metric for assessing model performance.

5.1 Dummy Classifier

To establish a baseline for evaluating model performance, a Dummy Classifier was first trained. This baseline model predicts the majority class ("Low Engagement") based purely on the distribution of the target variable. By comparing the results of more advanced models to this simple baseline, it becomes possible to assess how much predictive power is gained by incorporating textual, numerical, and categorical features, along with more sophisticated machine learning algorithms.

Evaluation Metric	Value
Accuracy on test set	0.75
Precision (negative-low engagement)	0.75
Recall (negative-low engagement)	1.00
F1-score (negative-low engagement)	0.86
Precision (positive-high engagement)	0.00
Recall (positive-high engagement)	0.00
F1-score (positive-high engagement)	0.00

Table 10 - Evaluation for Dummy Classifier (Source: author)

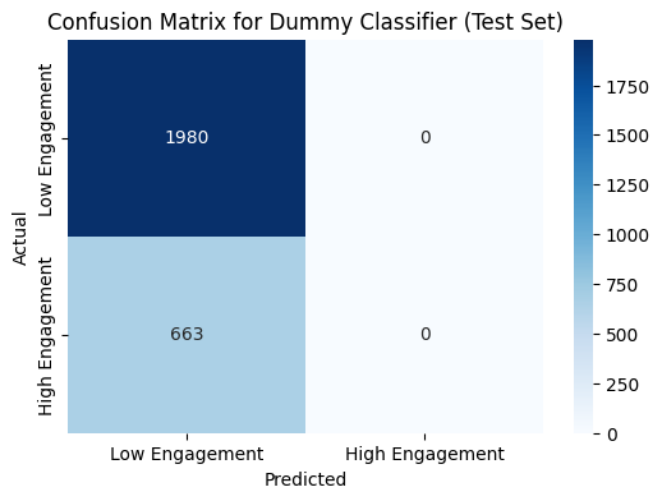


Figure 8 - Confusion Matrix for Dummy Classifier (Source: author)

As expected, the Dummy Classifier achieved relatively high overall accuracy (Table 10) because of the class imbalance, but it completely failed to identify instances of "High Engagement," as illustrated by the confusion matrix (Figure 8).

5.2 Logistic Regression

Evaluation Metric	Value
Accuracy on test set	0.72
Precision (negative-low engagement)	0.85
Recall (negative-low engagement)	0.76
F1-score (negative-low engagement)	0.80
Precision (positive-high engagement)	0.46
Recall (positive-high engagement)	0.59
F1-score (positive-high engagement)	0.51

Table 11 - Evaluation of Logistic Regression (Source: author)

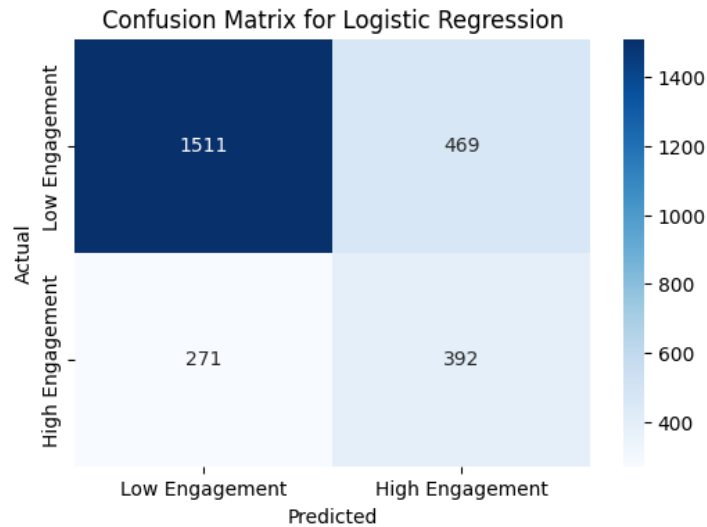


Figure 9 - Confusion Matrix for Logistic Regression (Source: author)

The confusion matrix shows that the Logistic Regression model correctly identified 1511 instances of low engagement and 392 instances of high engagement. However, it misclassified 469 low engagement articles as high engagement and 271 high engagement articles as low engagement.

5.3 Decision Trees

Evaluation Metric	Value
Accuracy on test set	0.71
Precision (negative-low engagement)	0.82
Recall (negative-low engagement)	0.80
F1-score (negative-low engagement)	0.81
Precision (positive-high engagement)	0.43
Recall (positive-high engagement)	0.46
F1-score (positive-high engagement)	0.45

Table 12 - Evaluation of Decision Trees (Source: author)

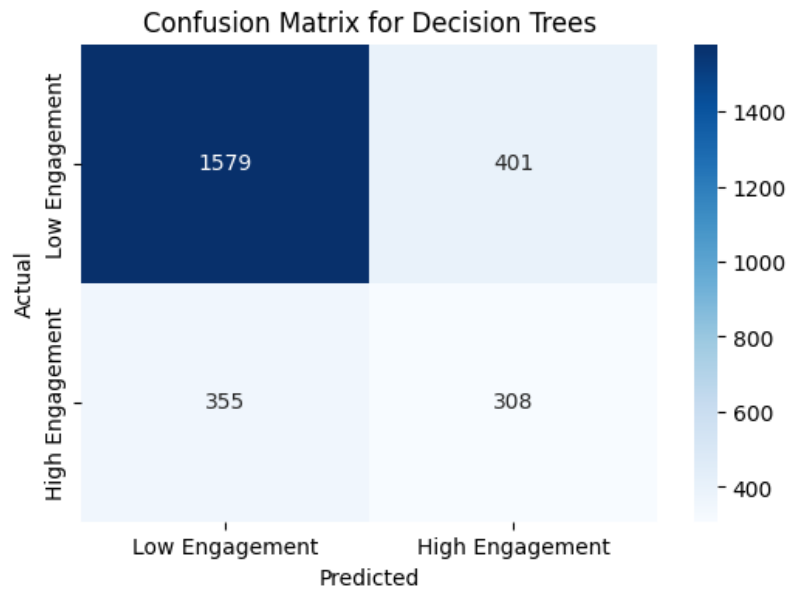


Figure 10 - Confusion Matrix for Decision Trees (Source: author)

The confusion matrix shows that the Decision Trees model correctly identified 1579 instances of low engagement and 308 instances of high engagement. However, it misclassified 401 low engagement articles as high engagement and 355 high engagement articles as low engagement.

5.4 Random Forest

Evaluation Metric	Value
Accuracy on test set	0.73
Precision (negative-low engagement)	0.84
Recall (negative-low engagement)	0.79
F1-score (negative-low engagement)	0.81
Precision (positive-high engagement)	0.47
Recall (positive-high engagement)	0.55
F1-score (positive-high engagement)	0.50

Table 13 - Evaluation of Random Forest (Source: author)

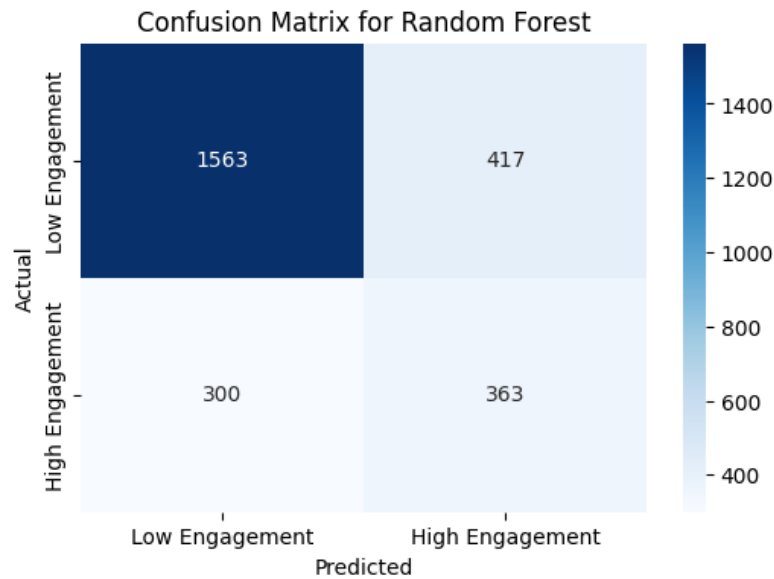


Figure 11 - Confusion Matrix for Random Forest (Source: author)

The confusion matrix shows that the Random Forest model correctly identified 1563 instances of low engagement and 363 instances of high engagement. However, it misclassified 417 low engagement articles as high engagement and 300 high engagement articles as low engagement.

5.5 Support Vector Machine

Evaluation Metric	Value
Accuracy on test set	0.73
Precision (negative-low engagement)	0.84
Recall (negative-low engagement)	0.78
F1-score (negative-low engagement)	0.81
Precision (positive-high engagement)	0.47
Recall (positive-high engagement)	0.57
F1-score (positive-high engagement)	0.51

Table 14 - Evaluation of SVM (Source: author)

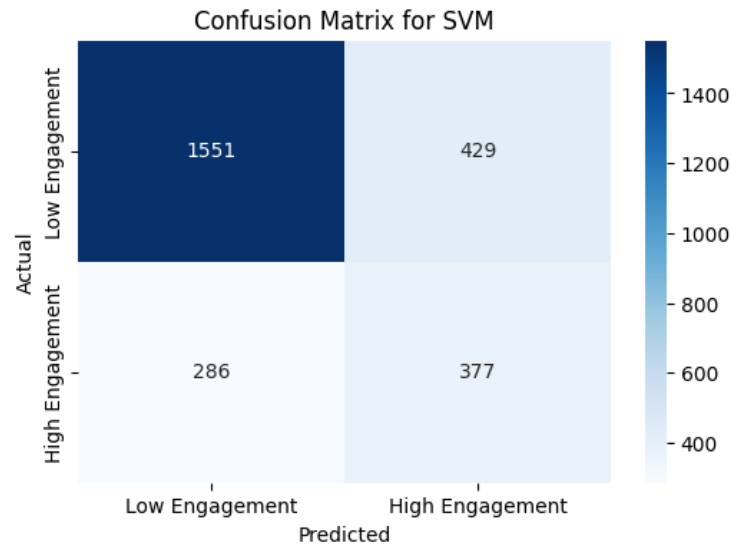


Figure 12 - Confusion Matrix of SVM (Source: author)

The confusion matrix shows that the SVM model correctly identified 1551 instances of low engagement and 377 instances of high engagement. However, it misclassified 429 low engagement articles as high engagement and 286 high engagement articles as low engagement.

5.6 XGBoost

Evaluation Metric	Value
Accuracy on test set	0.74
Precision (negative-low engagement)	0.84
Recall (negative-low engagement)	0.81
F1-score (negative-low engagement)	0.82
Precision (positive-high engagement)	0.48
Recall (positive-high engagement)	0.54
F1-score (positive-high engagement)	0.51

Table 15 - Evaluation of XGBoost (Source: author)

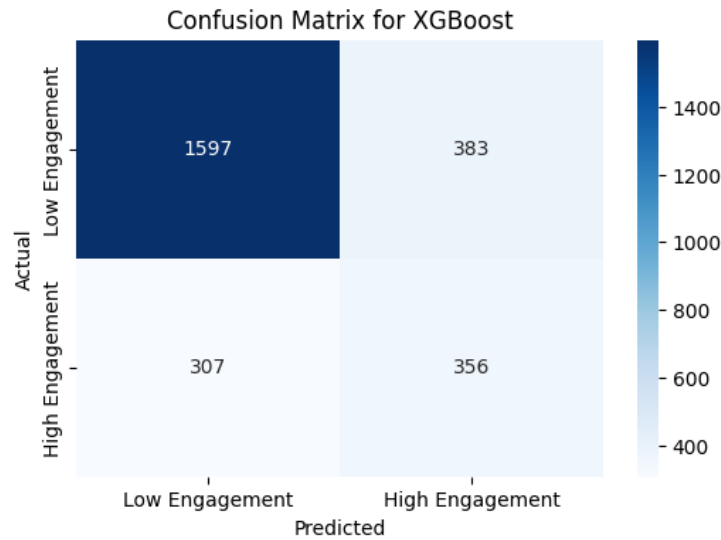


Figure 13 - Confusion Matrix of XGBoost (Source: author)

The confusion matrix shows that the XGBoost model correctly identified 1597 instances of low engagement and 356 instances of high engagement. However, it misclassified 383 low engagement articles as high engagement and 307 high engagement articles as low engagement.

5.7 Comparison of classifiers

When evaluating the performance of the classification models, particular attention was given to the F1-score for the high engagement class, as this class is both less frequent and more important for identifying impactful content.

The Dummy Classifier, used as a baseline, achieved a deceptively high overall accuracy due to the class imbalance, but completely failed to recognize any instances of high engagement, resulting in an F1-score of 0.00 for this class. This outcome confirms the need for more sophisticated models capable of distinguishing between the two categories.

Among the trained models, Logistic Regression, Support Vector Machine (SVM), and XGBoost all achieved the same F1-score of 0.51 for the high engagement class, demonstrating the best balance between precision and recall. Random Forest followed closely with an F1-score of 0.50, while the Decision Tree model performed the worst on this class, achieving an F1-score of 0.45.

Performance across all models was stronger on the low engagement class, with F1-scores ranging from 0.80 to 0.82. This difference reflects the influence of class imbalance, where predicting the majority class is less challenging.

In terms of recall, Logistic Regression achieved the highest recall for the high engagement class at 0.59, outperforming SVM (0.57) and XGBoost (0.54). This makes Logistic Regression the best at capturing high engagement articles among the models evaluated.

Figure 14 presents a comparison of all classifiers used in this experiment based on their F1-scores for both classes.

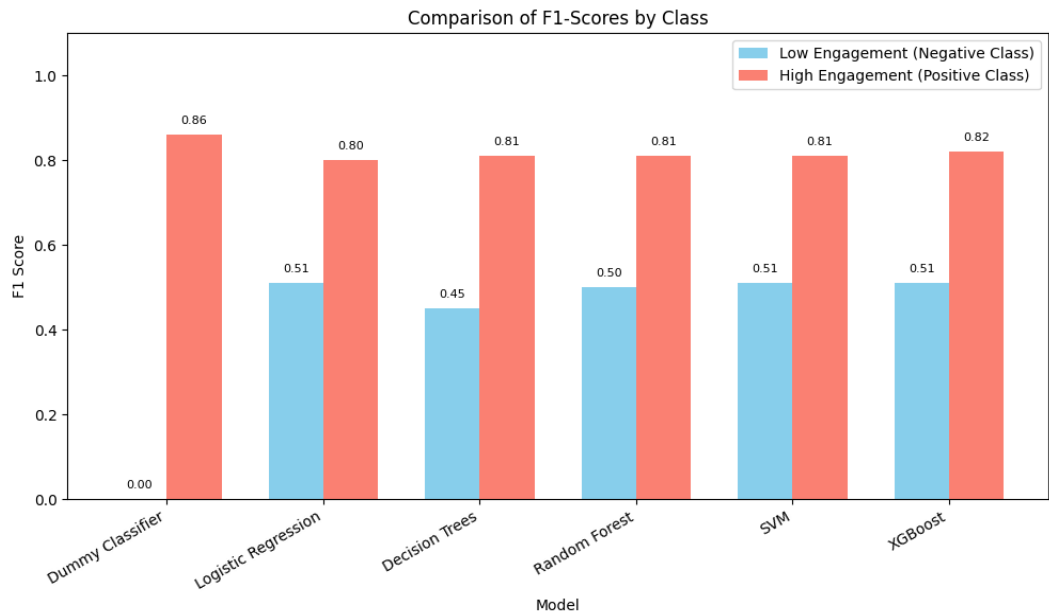


Figure 14 - Comparison of classifiers based on F1-scores (Source: author)

Figure 15 presents a comparison of all classifiers used in this experiment based on their recall for minority class.

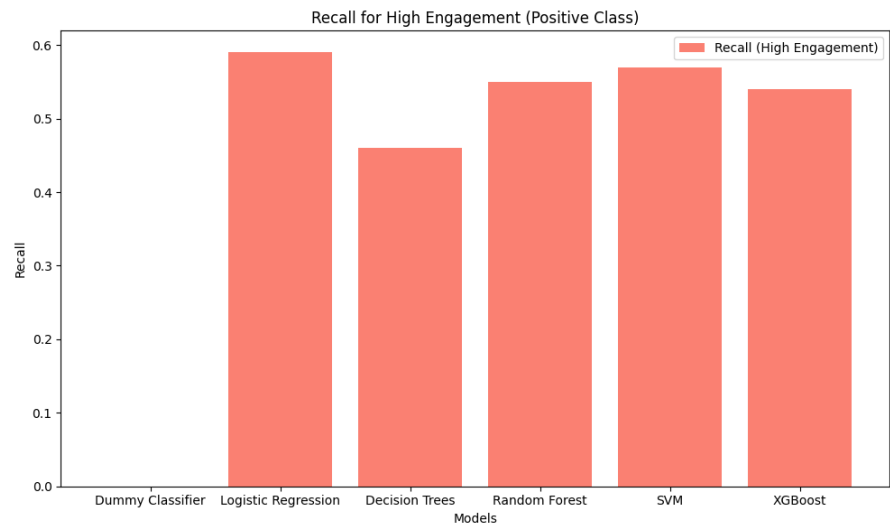


Figure 15 - Recall comparison for minority class (Source: author)

Conclusion

The main goal of this bachelor thesis was to create a model that predicts user engagement with news articles, focusing on key textual features such as headlines and teasers based on the articles from iDNES.cz news portal.

An analysis of feature importance further revealed that certain topics such as the war in Ukraine and Czech political figures were strong predictors of engagement. However, while these topics drive engagement, it is important that publishers approach them responsibly, avoiding the use of sensationalism or emotionally manipulative language.

The experiment faced several challenges. The most notable was class imbalance, with far fewer high engagement articles than low engagement ones, making it harder for models to correctly identify the more impactful content. Another limiting factor was the lack of advanced tools and resources for Czech natural language processing, which restricted the depth of preprocessing and analysis compared to what might be possible in English.

Among the models evaluated, Logistic Regression showed the best performance for predicting high engagement articles, with an F1-score of 0.51 and the highest recall for the high engagement class at 0.59. Support Vector Machine (SVM) and XGBoost both achieved an F1-score of 0.51 for high engagement, but their recall scores were lower than that of Logistic Regression, making them less effective in identifying high engagement articles. Decision Trees, on the other hand, performed the worst, with an F1-score of 0.45.

Low engagement articles were consistently easier for all models to predict, with F1-scores between 0.80 and 0.82. This is largely due to their higher representation in the dataset and the clearer patterns they presented.

Future work could expand on this study by incorporating additional features. Moreover, experimenting with deep learning models or fine-tuned language models like BERT might further improve performance. Addressing the class imbalance problem by trying more advanced techniques could enhance the model's ability to predict high engagement articles more effectively.

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